EXPERIMENT: 7

File System Forensic Analysis - Autopsy

DATE: 04/03/2026

AIM

To perform File System Forensics Analysis using Autopsy.

THEORY

File System Analysis Using Autopsy is a digital forensics platform and graphical interface to The Sleuth Kite and other digital forensics tools. Law enforcement, military, and corporate examiners use it to investigate what happened on a computer. You can even use it to recover photos from your camera's memory card. It can be directly used to find evidence.

- Volume and File System Analysis
- Plug-in Framework
- Download
- Documents
- History
- Licenses

Autopsy is a graphical interface to the command line digital investigation analysis tools in The Sleuth Kit. Together, they can analyze Windows and UNIX disks and file systems (NTFS, FAT, UFS1/2, Ext2/3).

The Sleuth Kit and Autopsy are both Open Source and run on UNIX platforms (you can use Cygwin to run them both on Windows). As Autopsy is HTML-based, you can connect to the Autopsy server from any platform using an HTML browser. Autopsy provides a "File Manager"-like interface and shows details about deleted data and file system structures.

Analysis Modes

- A dead analysis occurs when a dedicated analysis system is used to examine the data from a suspect system. In this case, Autopsy and The Sleuth Kit are run in a trusted environment, typically in a lab. Autopsy and TSK support raw, Expert Witness, and AFF file formats.
- A live analysis occurs when the suspect system is being analyzed while it is running. In this case, Autopsy and The Sleuth Kit are run from a CD in an untrusted environment. This is frequently used during incident response while the incident is being confirmed. After it is confirmed, the system can be acquired and a dead analysis performed.

Evidence Search Techniques

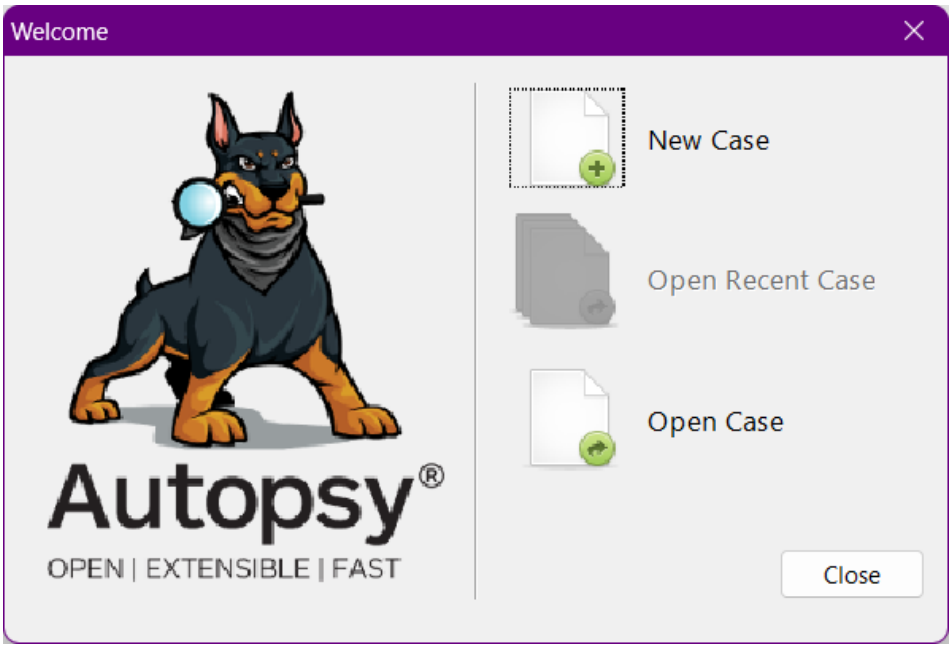
- File Listing: Analyze the files and directories, including the names of deleted files and files with Unicode-based names.
- File Content: The contents of files can be viewed in raw, hex, or the ASCII strings can be extracted. When data is interpreted, Autopsy sanitizes it to prevent damage to the local analysis system. Autopsy does not use any client-side scripting languages.
- Hash Databases: Lookup unknown files in a hash database to quickly identify it as good or bad. Autopsy uses the NIST National Software Reference Library (NSRL) and user created databases of known good and known bad files.
- File Type Sorting: Sort the files based on their internal signatures to identify files of a known type. Autopsy can also extract only graphic images (including thumbnails). The extension of the file will also be compared to the file type to identify files that may have had their extension changed to hide them.
- Timeline of File Activity: In some cases, having a timeline of file activity can help identify areas of a file system that may contain evidence. Autopsy can create timelines that contain entries for the Modified, Access, and Change (MAC) times of both allocated and unallocated files.
- Keyword Search: Keyword searches of the file system image can be performed using ASCII strings and grep regular expressions. Searches can be performed on either the full file system image or just the unallocated space. An index file can be created for faster searches. Strings that are frequently searched for can be easily configured into Autopsy for automated searching.
- Meta Data Analysis: Meta Data structures contain the details about files and directories. Autopsy allows you to view the details of any metadata structure in the file system. This is useful for recovering deleted content. Autopsy will search the directories to identify the full path of the file that has allocated the structure.
- Data Unit Analysis: Data Units are where the file content is stored. Autopsy allows you to view the contents of any data unit in a variety of formats including ASCII, hexdump, and strings. The file type is also given and Autopsy will search the metadata structures to identify which has allocated the data unit.
- Image Details: File system details can be viewed, including on-disk layout and times of activity. This mode provides information that is useful during data recovery.

Autopsy is written in Perl and runs on the same UNIX platforms as The Sleuth Kit:

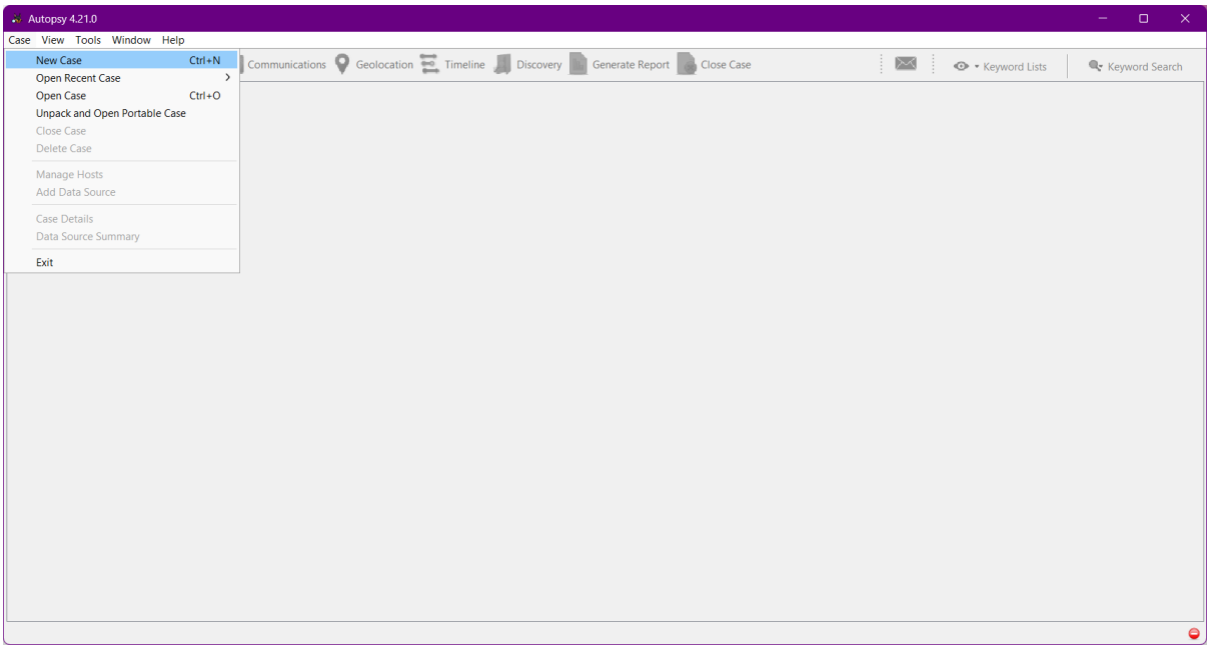
- Linux
- Mac OS X
- Open & FreeBSD
- Solaris
- Cygwin

PROCEDURE

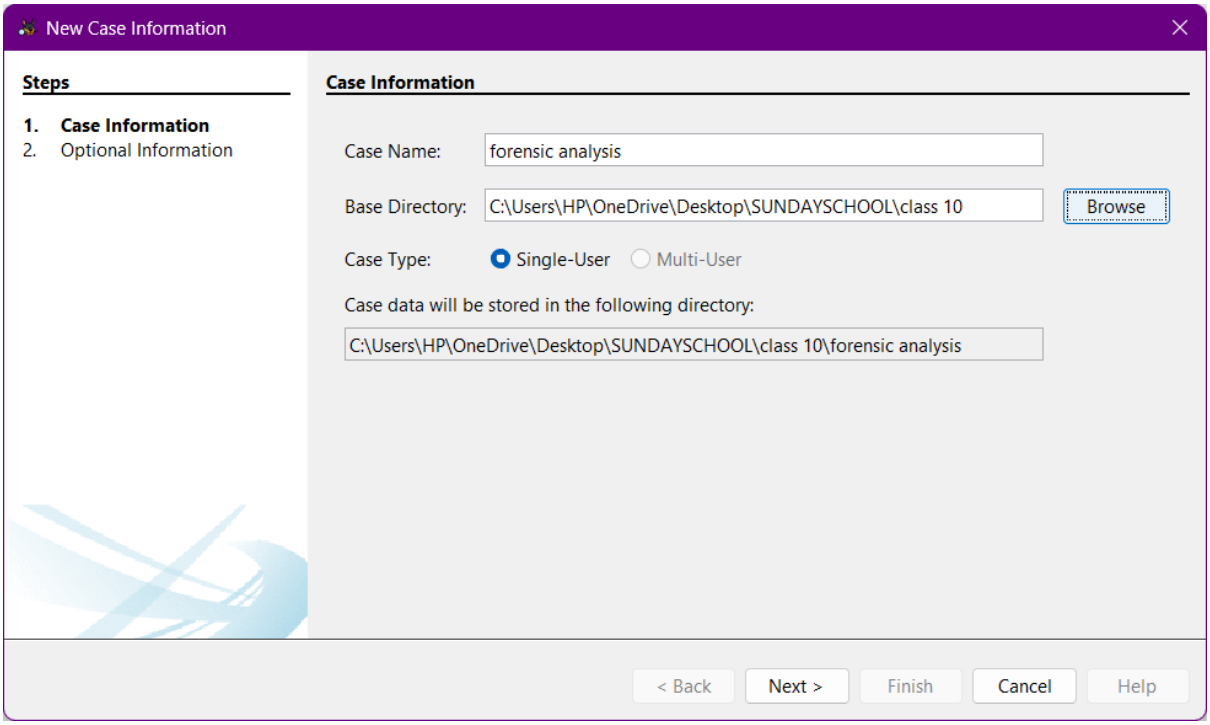
1.Run Autopsy and select New Case



Or

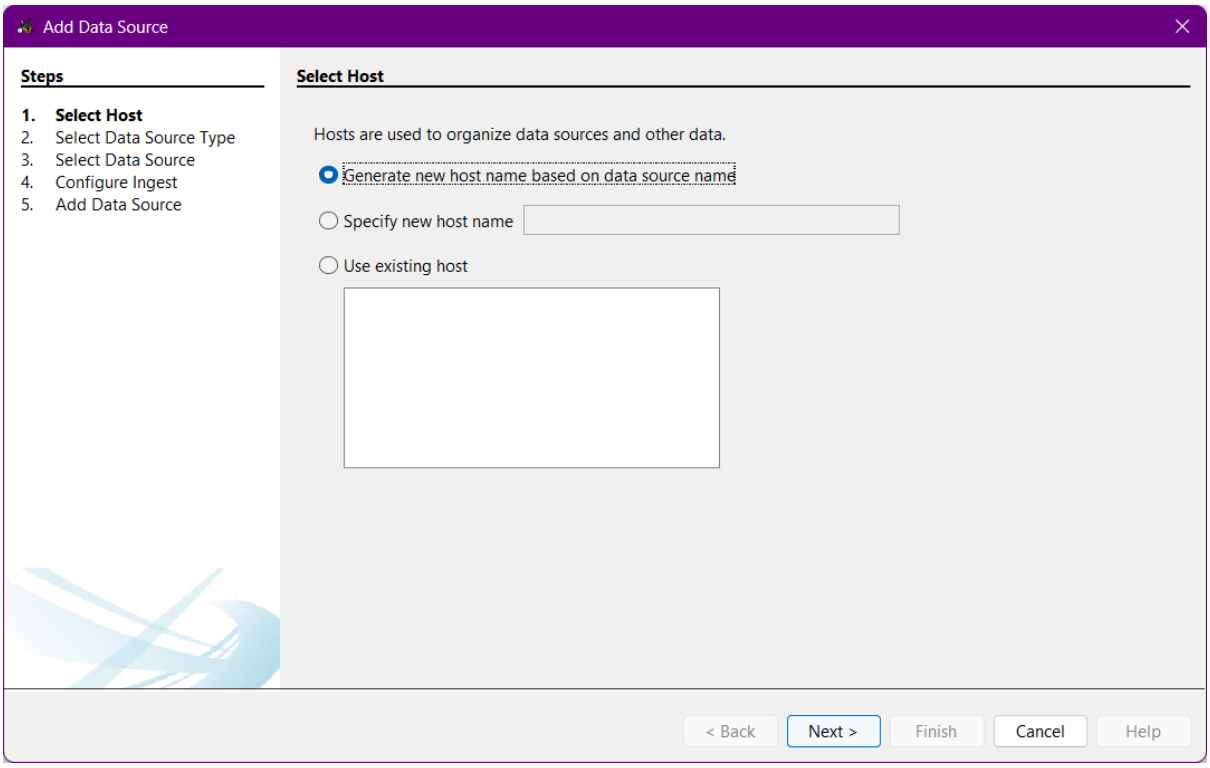


2.Provide the Case Name and the directory to store the case file. Click on Next.

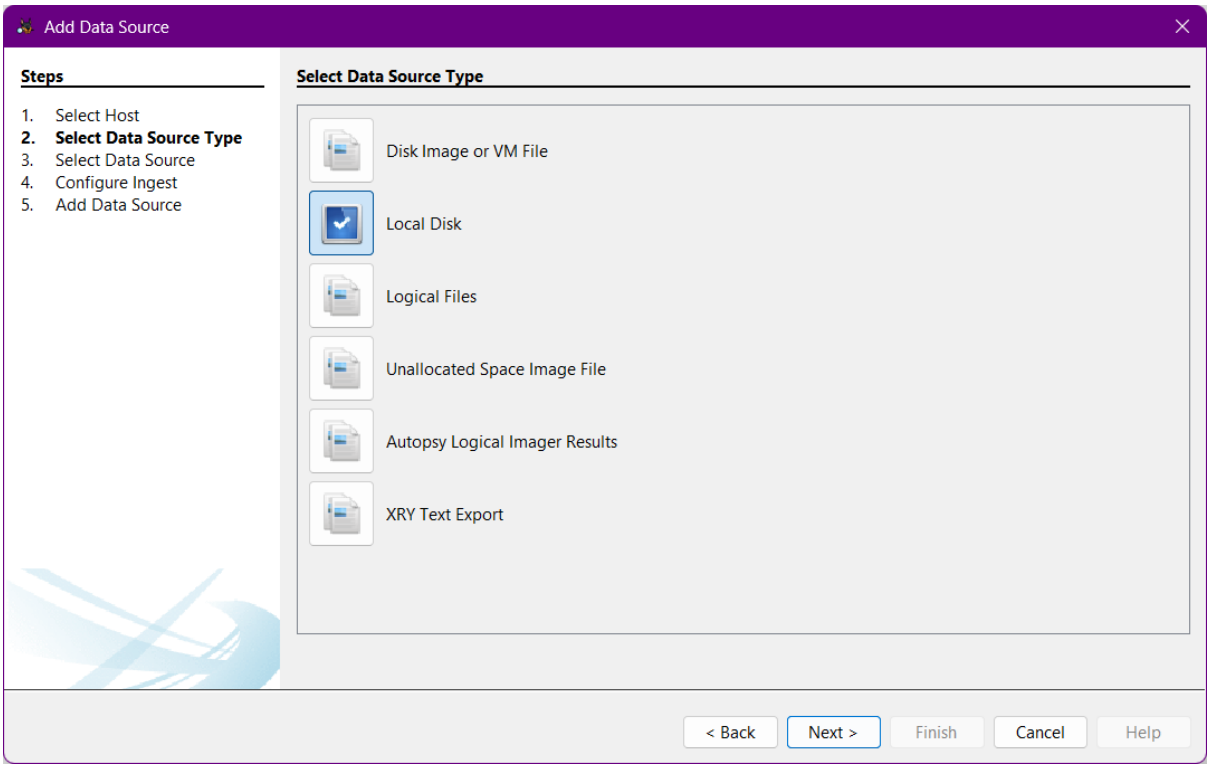


1. Add Case Number and Examiner's details, then click on Finish.

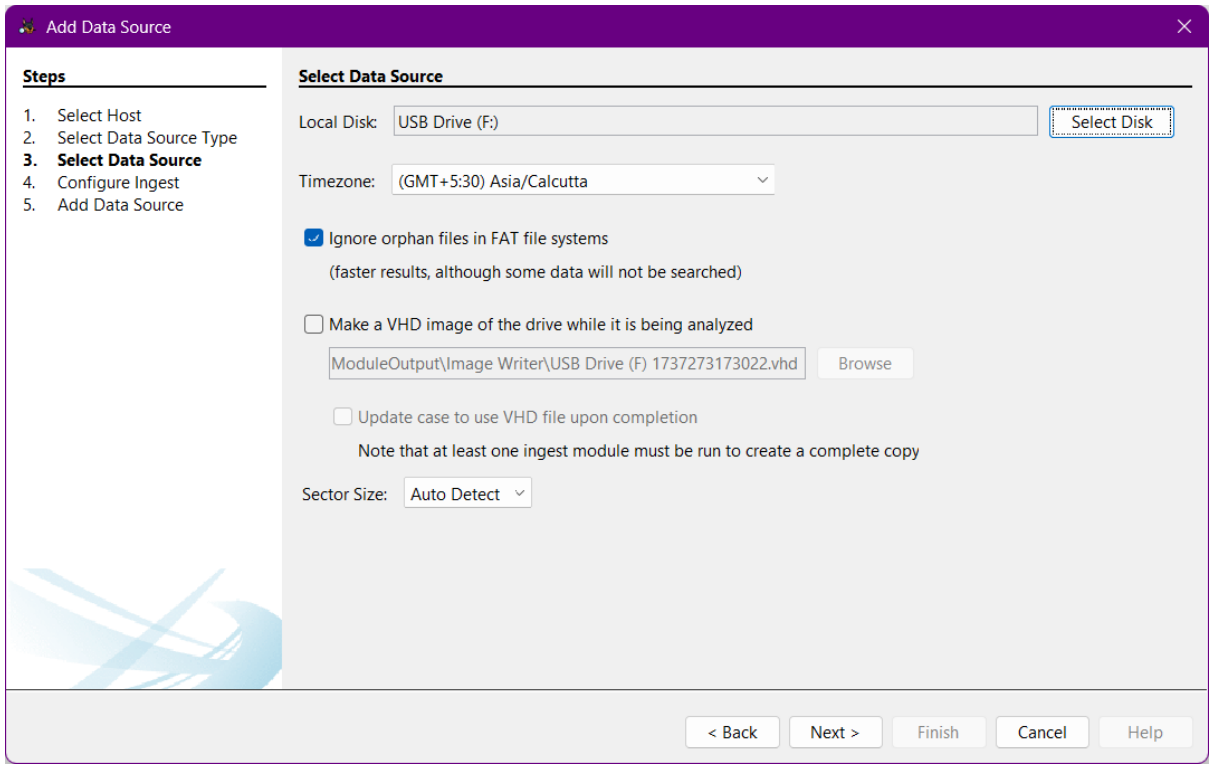
2. Select Host(Generate new host name based on data source name) and click Next.



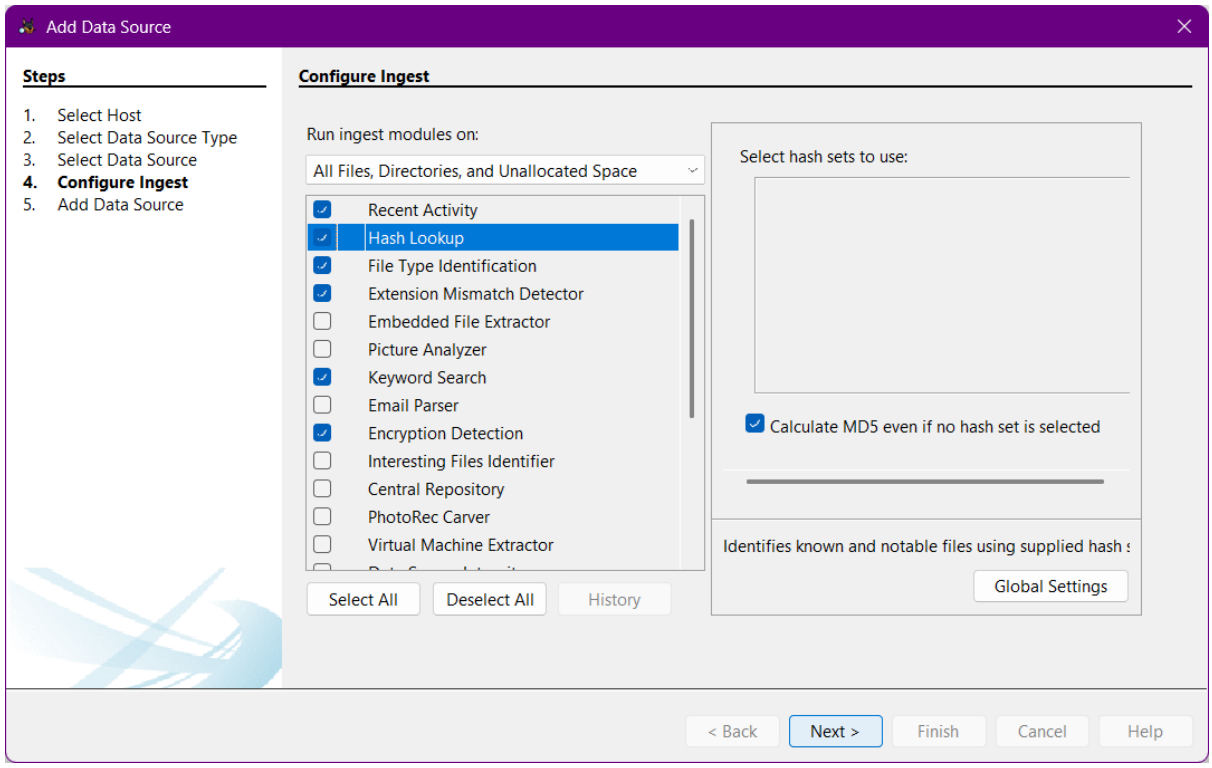
3. Choose the required data source type, in this case Disk Image and cick on Next.

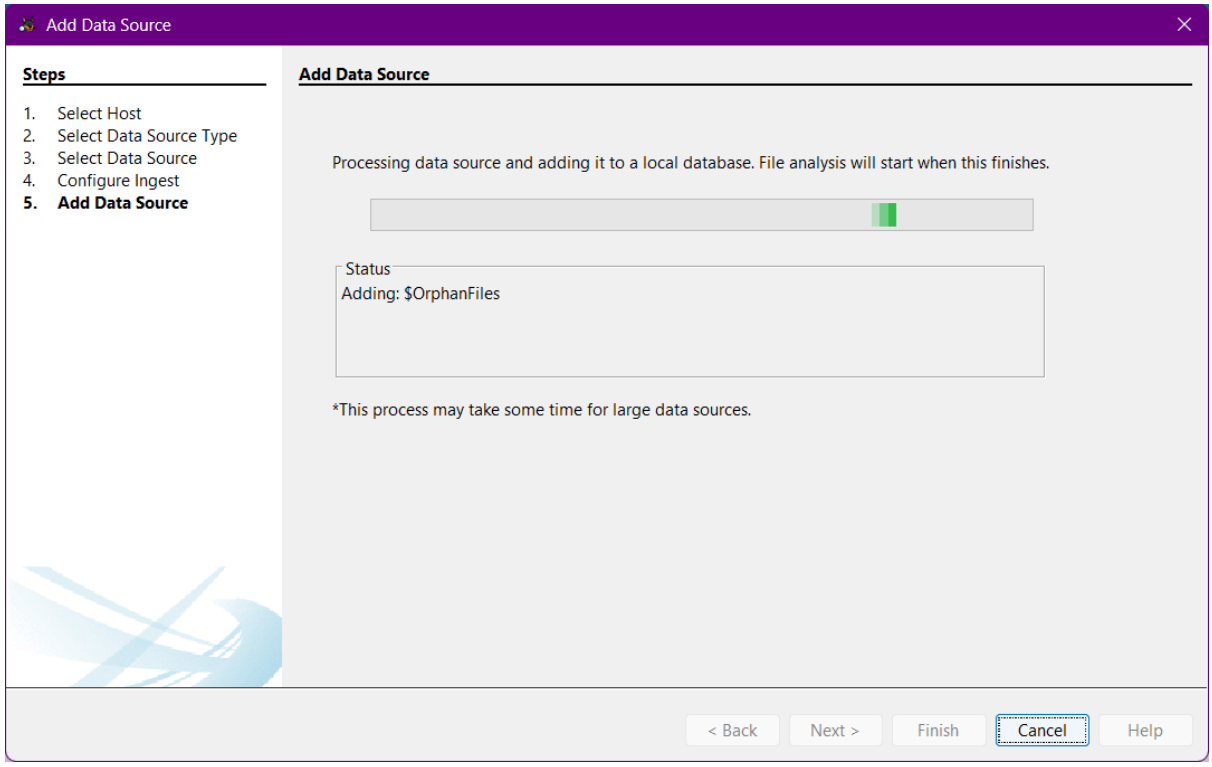


6.Give path of the data source(Select the local disk from the list) and make sure the Time zone is correct. Then click on next

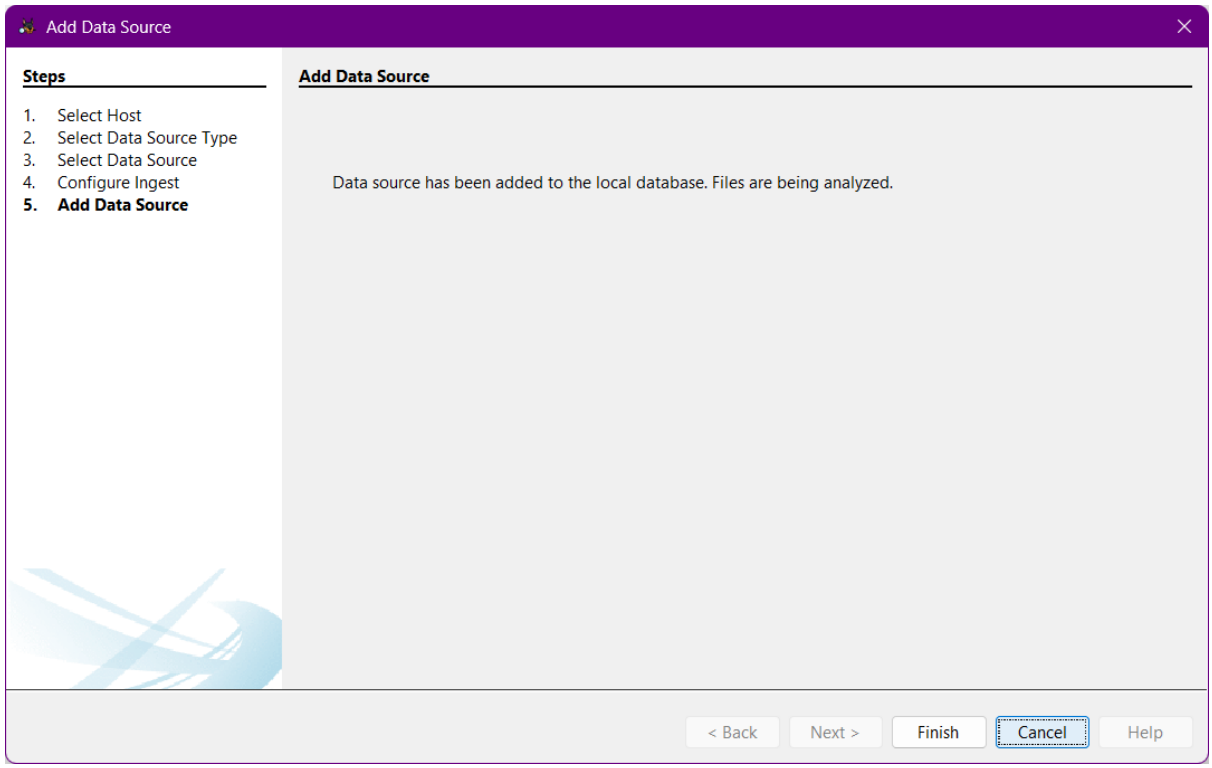


7.Configure Ingest. Select the actions we want to perform on selected disk and click on Next

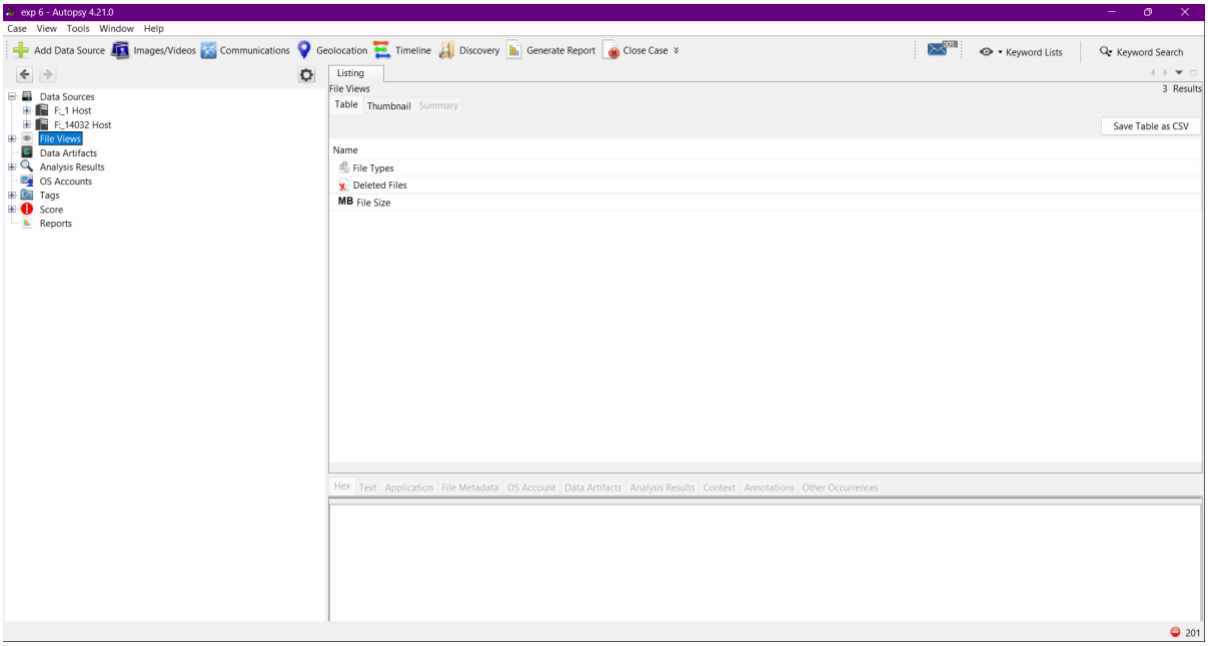

Add Data Source
Steps
1. Select Host
2. Select Data Source Type
3. Select Data Source
4. Configure Ingest
5. Add Data Source
Configure Ingest
Run ingest modules on:
All Files, Directories, and Unallocated Space
Recent Activity
Hash Lookup
File Type Identification
Extension Mismatch Detector
Embedded File Extractor
Picture Analyzer
Keyword Search
Email Parser
Encryption Detection
Interesting Files Identifier
Central Repository
PhotoRec Carver
Virtual Machine Extractor
Select All
Deselect All
History
Select hash sets to use:
Calculate MD5 even if no hash set is selected
Identifies known and notable files using supplied hash
Global Settings
< Back
Next >
Finish
Cancel
Help

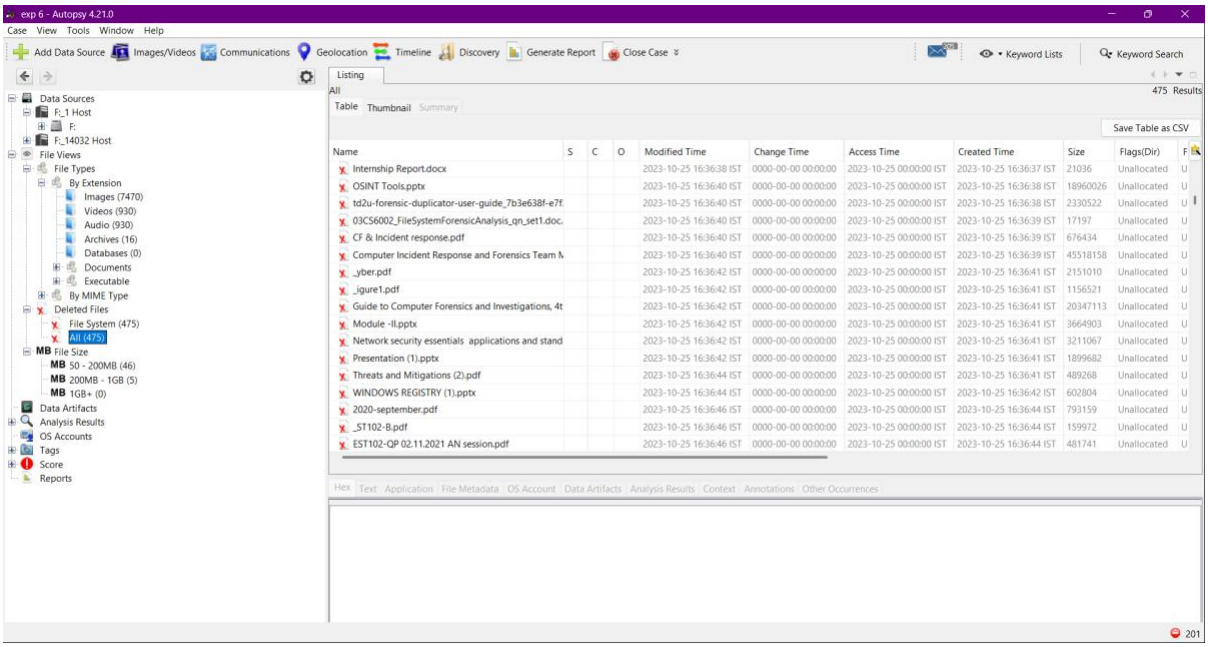

Add Data Source
Steps
1. Select Host
2. Select Data Source Type
3. Select Data Source
4. Configure Ingest
5. Add Data Source
Add Data Source
Processing data source and adding it to a local database. File analysis will start when this finishes.
Status
Adding: $OrphanFiles
*This process may take some time for large data sources.
< Back
Next >
Finish
Cancel
Help

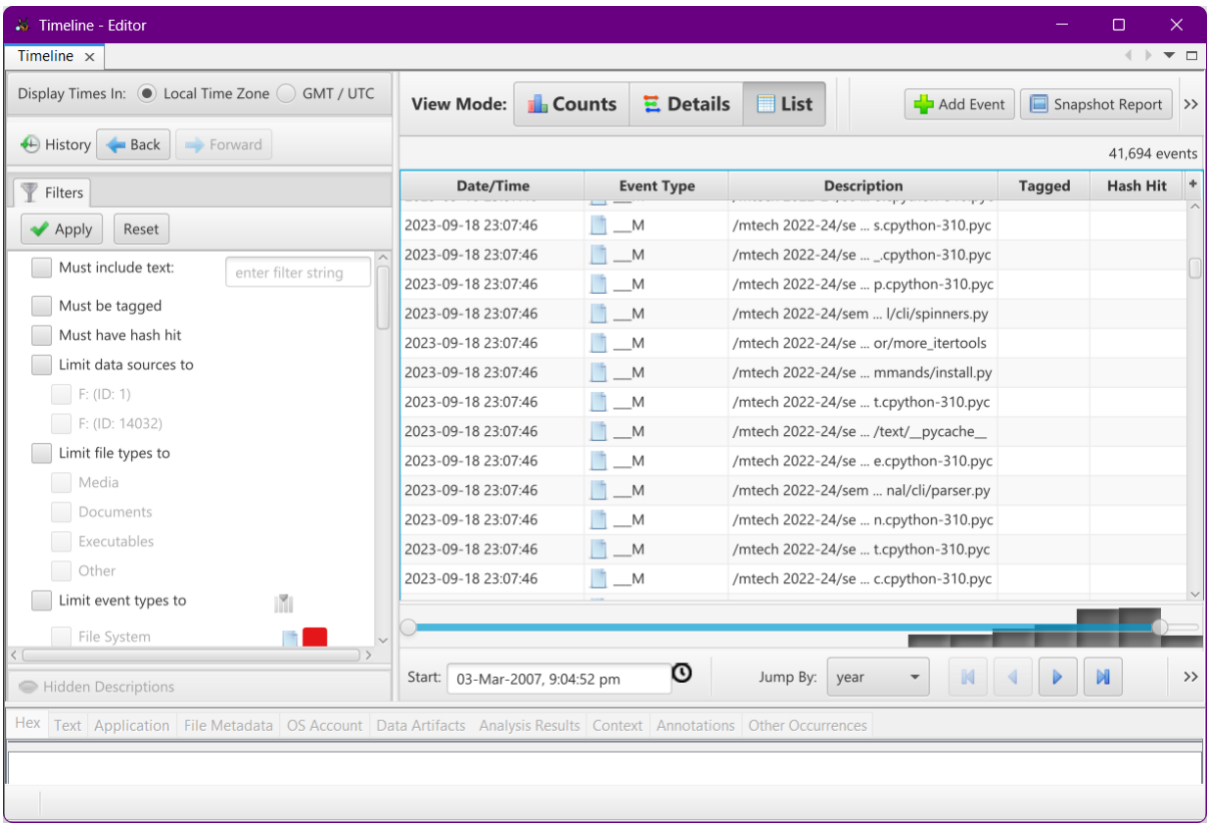
8.Data source has been added and click on Finish



Output







RESULT

File System Forensics Analysis using Autopsy is performed